

SAT Hard Question 20

1

★ Mark for Review

ABC

A computer program models the total mass of the population of a certain type of algae after the algae was placed in an environment where it has no natural predators. According to the model, the estimated total mass of this population of algae at the end of every 6-hour period is 129% greater than the estimated total mass of this population of algae at the end of the previous 6-hour period, and the estimated total mass of this population of algae is 613.90 grams after 18 hours. Which equation best represents this model, where A is the estimated total mass, in grams, of the population of algae after x hours, and $x < 50$?

A. $A = 34.11(1.29)^{\frac{x}{6}}$

B. $A = 34.11(2.29)^{\frac{x}{6}}$

C. $A = 51.12(2.29)^{\frac{x}{6}}$

D. $A = 285.98(1.29)^{\frac{x}{6}}$

2

★ Mark for Review

ABC

A solar panel is supported by two braces that form acute angles X and Y with two perpendicular sides of its frame. The measure of angle Y is 32° , and a load calculation for the braces gives $\frac{\sin X}{\cos Y} = 1$. What is the measure, in degrees, of angle X ?

A. 77

B. 58

C. 45

D. 32

3

★ Mark for Review

ABC

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(7x - 410)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-14x + w)^\circ$. What is the value of w ?

4

★ Mark for Review

ABC

The equation $M = 1,800(1.02)^t$ models the number of members, M , of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

A. $M = 1,800(1.02)^{\frac{q}{4}}$

B. $M = 1,800(1.02)^{4q}$

C. $M = 1,800(1.005)^{4q}$

D. $M = 1,800(1.082)^q$

5

★ Mark for Review

ABC

An engineer uses the equation

$$\frac{1}{24}x^2 + \left(s - \frac{1}{24}t\right)x - st = 0$$

to determine two possible signed calibration adjustments, x , for a machine. In the equation, s and t are positive constants. The product of the two possible adjustments is $-2kst$, where k is a constant. What is the value of k ?

6

★ Mark for Review

ABC

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y , in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 1.75 inches, it has an estimated breaking strength of 3,858.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 3.50 inches?

7

★ Mark for Review

ABC

A hollow steel pipe is in the shape of a right circular cylinder. The steel pipe has an outside diameter of 72 inches, a wall thickness of $\frac{5}{8}$ inches, and a height of 138 inches. Which of the following is closest to the volume, in cubic inches, of the wall of this steel pipe?

A. 169

B. 542

C. 3,105

D. 19,340

8

★ Mark for Review

ABC

Trapezoid $CDEF$ is similar to trapezoid $JKLM$, such that C, D, E , and F correspond to J, K, L , and M , respectively. The length of each side of trapezoid $JKLM$ is 4 times the length of its corresponding side in trapezoid $CDEF$. The measure of angle C is 42° . What is the measure of angle J ?

A. 38° B. 42° C. 46° D. 168° **9**

★ Mark for Review

ABC

A robotic arm rotates through an angle F of $\frac{\pi}{15}$ radians. Its control panel displays this rotation as $6n$ degrees, where n is a constant. What is the value of n ?

10

★ Mark for Review

ABC

The table gives the average time t , in minutes, it takes Adriana to travel a certain distance d , in miles. Which equation could represent this linear relationship?

Distance (miles)	Average time (minutes)
0.18	9
0.24	12
0.36	18

A. $t = 50d$

B. $t = \frac{1}{2}d$

C. $t = 2d$

D. $t = \frac{1}{50}d$

11

★ Mark for Review

ABC

A wireless phone company pays customers for trading in old smartphones. The payment p , in dollars, is the original retail price of a smartphone minus \$135 for each year since the smartphone was released. Which equation represents this situation for a smartphone with an original retail price of \$750 when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

A. $p = 135 - 750t$

B. $p = 750 - 135t$

C. $t = 135 - 750p$

D. $t = 750 - 135p$

12

★ Mark for Review

ABC

A soccer team's goal is to earn at least \$1,170 by selling coupon books. The team earns \$9 from selling each coupon book. Which of the following inequalities describes all possible values for the number of coupon books, n , the team can sell to meet the goal?

A. $n + 9 \leq 1,170$

B. $n + 9 \geq 1,170$

C. $9n \leq 1,170$

D. $9n \geq 1,170$

13

★ Mark for Review

ABC

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 62 sedans for sale at the dealership on Monday, how many minivans were for sale?

14

★ Mark for Review

ABC

A snack company advertises that bags of pretzels produced by the company weigh, on average, 1 pound each. To test this, Sam selected at random 30 bags of pretzels produced by the company and weighed each bag. Based on his measurements, Sam estimated that for all bags of pretzels produced by the company, the average weight per bag is 1.02 pounds, with a margin of error of 0.05 pounds. For all bags of pretzels produced by the company, which of the following is the most appropriate conclusion about the average weight w , in pounds, per bag?

A. $w = 1.02$

B. $0.97 \leq w \leq 1.02$

C. $0.97 \leq w \leq 1.07$

D. $1.02 \leq w \leq 1.07$

15

★ Mark for Review

ABC

A total of 3 equilateral pentagons each have side length p units. A total of 2 squares each have side length s units. The sum of the perimeters of all these pentagons and squares is 136 units. Which equation represents this situation?

A. $12p + 10s = 136$

B. $2p + 3s = 136$

C. $3p + 2s = 136$

D. $15p + 8s = 136$

16

★ Mark for Review

ABC

A company conducted a study concerning the time spent on extracurricular activities by students in grades 6, 7, and 8 and their level of school spirit. A total of 80 students were selected at random from all students in grades 6, 7, and 8 in a large school district in the US. The selected students were asked to report their level of school spirit as either low, medium, or high. The table summarizes the distribution of the selected students by grade.

Grade	6	7	8
Number of students	26	25	29

If the results of the study suggest a relationship between the self-reported level of school spirit and the amount of time spent on extracurricular activities by students in grades 6, 7, and 8, which of the following aspects of the study would prevent the results from being generalized to all students in grades 6, 7, and 8 in the US?

A. The selected students are not all from the same school.

B. The number of selected students is too small.

C. The number of selected students from each grade is different.

D. The students were not selected at random from all students in grades 6, 7, and 8 in the US.

17

★ Mark for Review

ABC

For a study, a research team measured the heights of 240 king penguins from the Crozet Islands and 120 king penguins from the Falkland Islands. The research team determined that the mean height of the 240 king penguins from the Crozet Islands was 86 centimeters and the mean height of the 120 king penguins from the Falkland Islands was 92 centimeters. What was the mean height, in centimeters, of all 360 king penguins the research team measured for this study?

A. 87

B. 88

C. 89

D. 90

18

★ Mark for Review

ABC

At a nature preserve, a wildlife biologist counted bald eagles from an observation deck at the same time each day for 21 days. The table summarizes the resulting data set, data set A.

Number of bald eagles	Number of days
0	1
1	3
2	4
3	5
4	4
5	3
19	1

The data value 19 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 20 data values. Which statement best compares the median of data set A and the median of data set B?

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is equal to the median of data set A.
- C. The median of data set B is greater than the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

19

★ Mark for Review

ABC

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B.

Right circular cylinder A	392π
Right circular cylinder B	$10,584\pi$

The radius of right circular cylinder A is 7 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$?

(The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

A. 1008**B.** 1210**C.** 1680**D.** 2014

Q1: C	Q2: B	Q3: 1360	Q4: A
Q5: 12	Q6: 15,435	Q7: D	Q8: B
Q9: 2	Q10: A	Q11: B	Q12: D
Q13: 93	Q14: C	Q15: D	Q16: D
Q17: B	Q18: B	Q19: C	

Step-by-Step Solutions

Q1. A quantity that is 129% greater is $100\% + 129\% = 229\%$ of the previous quantity, so the 6-hour growth factor is 2.29. After 18 hours, there have been $18/6 = 3$ growth periods. If A_0 is the initial mass, then

$$613.90 = A_0(2.29)^3, \quad A_0 = \frac{613.90}{(2.29)^3} \approx 51.12.$$

Thus $A = 51.12(2.29)^{x/6}$. **Answer: C.**

Q2. Since $\frac{\sin X}{\cos Y} = 1$, we have $\sin X = \cos Y$. Use the cofunction identity $\cos Y = \sin(90^\circ - Y)$:

$$\sin X = \sin(90^\circ - 32^\circ) = \sin 58^\circ.$$

Both X and Y are acute, so $X = 58^\circ$. **Answer: B.**

Q3. Let the acute angle be $a = 7x - 410$. Each obtuse angle is supplementary to it, so its measure is $180 - a$. The stated sum is

$$a + 3(180 - a) = 540 - 2a.$$

Substitute $a = 7x - 410$:

$$540 - 2(7x - 410) = 540 - 14x + 820 = -14x + 1360.$$

Comparing this with $-14x + w$ gives $w = 1360$. **Answer: 1360.**

Q4. Since one year contains 4 quarter years, q quarter years is $q/4$ years. Substitute $t = q/4$ into the original model:

$$M = 1,800(1.02)^{q/4}.$$

Answer: A.

Q5. For $ax^2 + bx + c = 0$, the product of the solutions is c/a . Here, $a = 1/24$ and $c = -st$, so

$$\frac{c}{a} = \frac{-st}{1/24} = -24st.$$

Set this equal to the given product:

$$-24st = -2kst.$$

Because s and t are positive, divide both sides by $-2st$ to get $k = 12$. **Answer:** 12.

Q6. First use the known rope to find a :

$$3,858.75 = 900a(1.75)^2.$$

Since $900(1.75)^2 = 2,756.25$, $a = 3,858.75/2,756.25 = 1.4$. For $x = 3.50$,

$$y = 900(1.4)(3.50)^2 = 1,260(12.25) = 15,435.$$

Answer: 15,435.

Q7. The outside radius is $R = 72/2 = 36$ inches. The inside radius is

$$r = 36 - \frac{5}{8} = 35.375 \text{ inches.}$$

The wall volume is the outer-cylinder volume minus the inner-cylinder volume:

$$V = \pi(138)(36^2 - 35.375^2) \approx 19,339.94.$$

The closest choice is 19,340. **Answer:** D.

Q8. Similar figures have congruent corresponding angles. Since angle C corresponds to angle J ,

$$m\angle J = m\angle C = 42^\circ.$$

The side-length scale factor does not change angle measures. **Answer:** B.

Q9. Convert radians to degrees:

$$\frac{\pi}{15} \left(\frac{180^\circ}{\pi} \right) = 12^\circ.$$

Since the display reads $6n$ degrees, $6n = 12$, so $n = 2$. **Answer:** 2.

Q10. Compute the constant ratio using any row:

$$\frac{t}{d} = \frac{9}{0.18} = 50.$$

The other rows give the same ratio, so $t = 50d$. **Answer:** A.

Q11. The phone starts at \$750 and loses \$135 in value for each of t years. Therefore,

$$p = 750 - 135t.$$

Answer: B.

Q12. Selling n books at \$9 earned per book produces $9n$ dollars. “At least” means greater than or equal to, so

$$9n \geq 1,170.$$

Answer: D.

Q13. The 62 sedans are 20% of all vehicles. If T is the total number, then

$$0.20T = 62 \quad \Rightarrow \quad T = 310.$$

Minivans make up $100\% - 20\% - 50\% = 30\%$ of the vehicles. Thus the number of minivans is

$$0.30(310) = 93.$$

Answer: 93.

Q14. The margin of error gives an interval centered at 1.02:

$$1.02 - 0.05 \leq w \leq 1.02 + 0.05.$$

Therefore, $0.97 \leq w \leq 1.07$. **Answer: C.**

Q15. One pentagon has perimeter $5p$, so three have total perimeter $3(5p) = 15p$. One square has perimeter $4s$, so two have total perimeter $2(4s) = 8s$. Hence

$$15p + 8s = 136.$$

Answer: D.

Q16. The students were randomly selected only from one large school district, not from all US students in grades 6, 7, and 8. Therefore, the sample may not represent the entire US population of those grades. **Answer: D.**

Q17. Use a weighted mean because the groups have different sizes:

$$\frac{240(86) + 120(92)}{240 + 120} = \frac{20,640 + 11,040}{360} = \frac{31,680}{360} = 88.$$

Answer: B.

Q18. Data set A has 21 values, so its median is the 11th value. Positions 9 through 13 are all 3, so its median is 3. After removing 19, data set B has 20 values; its median is the average of the 10th and 11th values. Both are 3, so its median is also 3. The medians are equal. **Answer: B.**

Q19. The volume ratio is

$$\frac{10,584\pi}{392\pi} = 27.$$

For similar solids, the volume ratio is the cube of the linear scale factor, so the linear scale factor is $\sqrt[3]{27} = 3$. For cylinder A,

$$392\pi = \pi(7^2)h \Rightarrow h = 8.$$

Thus its surface area is

$$2\pi(7^2) + 2\pi(7)(8) = 210\pi,$$

so $k = 210$. Surface area scales by the square of the linear factor, so $n = 3^2(210) = 1890$. Therefore,

$$n - k = 1890 - 210 = 1680.$$

Answer: C.